

Predicting Endosomal Escape: A Simulation-Based Analysis of pH-Responsive Lipid Nanoparticle Behavior

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Northampton Community College Academic Conference (Fall 2026)

Introduction

The rapid development of Messenger RNA (mRNA) therapeutics has advanced the techniques for treating genetic disorders and viral infections. However, many failures in mRNA engineering stem from particular limitations of Lipid Nanoparticles (LNPs). These synthetic nanoparticles are the primary carriers that protect mRNA from the unfavorable conditions of the bloodstream. An LNP's core function is to shield the mRNA cargo from degradation by specific enzymes called serum ribonucleases (RNases) once injected into the bloodstream. During in vitro storage of vaccine vials, such as the Pfizer-BioNTech COVID-19 mRNA vaccine, nucleophilic attack must be suppressed to slow the kinetics of alkaline hydrolysis. The threat of degradation continues in vivo. Scientists have discovered that to combat both threats, the easily breakable 2'OH groups attached to the ribose of the mRNA structure can be replaced with modified ether groups. The resulting ether linkage, such as 2'-O-Methyl, forms stronger bonds and is less likely to be severed from the phosphodiester backbone. These biological mechanisms are conditions that naked mRNA cannot withstand on its own. Modern LNP formulations integrate cationic ionizable lipids (CILs), cholesterol, polyethylene glycol lipids (PEG-lipids), and helper lipids to facilitate cellular uptake into target cells. The essential function of CILs is to reduce toxicity during circulation and enable cellular internalization. PEGylated lipids have a hydrophilic coating that determines the size of the LNPs; the more PEG lipids within the solution, the smaller the nanoparticle. This hydrophilic shielding effectively hides the LNP from the immune system and even helps maintain vial stability by preventing aggregation. Cholesterol and helper phospholipids strengthen the structure of the lipid bilayer by filling gaps between molecules and act as buffers to maintain fluidity at cold temperatures. Although these enhancements have established a protective shield around the mRNA, a critical obstacle remains: achieving reliable, timely endosomal escape.

As the LNP reaches the last stage, receptor-mediated endocytosis, the mature endosome's proton pumps acidify the vesicle, lowering the internal pH from the physiological 7.4 to approximately 5.5. In an acidic environment, the LNPs' CIL amine head groups are forced to shift from their neutral state to a positively charged state through protonation. This abrupt cationization triggers LNP to rupture the endosomal membrane

and release mRNA into the cytoplasm before lysosomal fusion digests the contents. Although this ideal scenario can be compromised if the CILs' pKa is too high, premature protonation in the early endosome, before the late endosome acidifies, leads to structural disruption and mRNA leakage. At the opposite end, destabilization can be delayed past the late endosomal window, and lysosomal fusion degrades the nucleic acid material.

Biopharmaceutical teams must therefore navigate a narrow window of membrane destabilization to ensure that the genetic instructions reach ribosomes and drive the synthesis of therapeutic proteins.

This project proposes a computational framework to analyze published molecular dynamics (MD) simulation trajectory data, to characterize how the LNP structure responds to an internal pH shift. By examining existing simulation datasets across a range of lipid molar compositions, the results can identify which formulations most reliably trigger the cylindrical-to-inverted hexagonal (H_{II}) phase transition. Since the lipid fractions examined are the same parameters that LNP formulation scientists adjust in the laboratory, the results can directly guide real-world experimental decisions as opposed to remaining purely theoretical. I am motivated by the idea that subtle changes in a fragile nucleic acid's environment can lead to life-saving therapies that safely reach the patients who need them most. These outputs will enable accurate predictions for the designs of next-generation mRNA delivery systems, with relevance to both clinically viable gene therapies and preventative vaccine development.

Specific Aims

Although LNPs are widely regarded as the leading delivery platform for mRNA therapeutics, their clinical efficacy remains fundamentally constrained by inefficient endosomal escape. Fewer than 3% of endocytosed LNPs successfully release their mRNA into the cytoplasm in early-generation formulations. The lamellar (L_{α})-to- H_{II} phase transition of CILs has not been fully resolved at the atomic level and is not yet predictable from lipid composition alone. LNP formulation remains a costly and time-intensive process in modern therapeutic development, with no clear understanding of how lipid molar ratios shift the pH threshold for targeted membrane rupture by the CILs.

This project pursues the following three aims:

Specific Aim 1: Validate published MD simulation trajectories of four-component LNP systems across physiological and endosomal pH environments.

The foundation of this investigation is the systematic curation of openly accessible coarse-grained MD simulation trajectories for four-component LNP systems. The research material will be sourced from the CHARMM-GUI membrane archive and the supplementary materials of peer-reviewed publications. Trajectories will be selected using the MARTINI coarse-grained force field, with SM-102, used in the Moderna COVID-19 mRNA vaccine, as the reference CIL. Where SM-102 parameters are unavailable, a well-documented analog, MC3, will be substituted. Selected datasets will cover three pH conditions: physiological (7.4), early endosomal (6.5), and late endosomal (5.5). Structural outputs, including membrane curvature, CIL lipid tail orientation, and water penetration depth, will be compared with published cryo-electron microscopy data and experimental encapsulation efficiency curves. This aim establishes the verified analytical dataset upon which Aims 2 and 3 depend.

Specific Aim 2: Characterize the protonation-dependent structural transitions governing endosomal membrane destabilization across varied lipid molar compositions.

Using the curated trajectory datasets from Aim 1, the goal will be to systematically analyze trajectories of the molar fractions of CIL and cholesterol, the two components most directly implicated in H_{II} phase formation. The outcomes will vary across a defined compositional range cross-referenced with clinically established formulations, including Onpattro and Moderna's mRNA-1273. For each composition, the analysis will track the onset and progression of the L_{α} to H_{II} phase transition. I will identify which lipid molar ratios most reliably trigger membrane rupture within the late endosomal window, before lysosomal fusion occurs. The N/P ratio for each lipid system represented in the curated datasets will be calculated and reported as a compositional output, reflecting the charge balance between CIL amine groups and the mRNA phosphate backbone.

Specific Aim 3: Identify optimal lipid molar compositions for endosomal escape efficiency and translate findings into experimentally actionable formulation guidelines.

Drawing on the structural transition data from Aim 2, the final objective will be to rank candidate lipid systems by their predicted endosomal escape efficiency. The metrics will be based on the speed and completeness of the L_{α} to H_{II} phase transition within the late endosomal pH window without premature destabilization in the early endosome. The CIL pKa range associated with optimal escape timing will be characterized, providing criteria for future CIL synthesis efforts. The results will reveal recommended ranges of molar ratios and CIL pKa targets. The goal is to provide formulation scientists with a clearer starting point, grounded in simulation data rather than prolonged trial and error.

Methodology

This study creates a computational framework to investigate LNP structural behavior across the three biologically relevant pH environments using publicly available MD simulation trajectory data from peer-reviewed publications. Instead of generating new simulation runs, this project will focus on existing LNP simulation datasets to systematically analyze and visualize the compositional transitions that dictate efficient mRNA release.

Confirmational analysis will be performed using MDAnalysis, an open-source Python library widely used in computational biophysics, to extract quantitative measurements from published simulation trajectories. Primary outputs will be evaluated across pH conditions that drive endosomal membrane rupture. The molecular visualization software VMD (Visual Molecular Dynamics) will generate high-resolution structural images of LNP behavior at each pH level.

Lipid molar compositions from three clinically validated LNP systems, Onpattro, Moderna mRNA-1273, and Pfizer-BioNTech BNT162b2, will be compiled and compared with their reported encapsulation and delivery outcomes in the literature. This compositional analysis will contextualize the simulation findings within real-world biopharmaceutical formulation practice and identify which molar-ratio parameters most consistently correlate with efficient endosomal escape. The N/P ratio for each clinical system will be reported as a derived characteristic of its lipid composition. Simulation datasets will be sourced from publicly available repositories, as described in Aim 1. Physical outputs will be validated by comparing measured parameters against published cryo-EM imaging data of LNP systems under equivalent pH conditions.

Expected Outcomes

The structural analysis is expected to produce quantifiable differences in membrane curvature, CIL lipid tail splay angle, and water penetration depth across the three pH environments, providing visual and numerical evidence of the progressive L_{α} to H_{II} phase transition that regulates endosomal membrane rupture. These insights are anticipated to reveal which compositional parameters most consistently correlate with efficient escape from late endosomes. Should results deviate from these expectations, the discrepancies will be recorded and used to isolate gaps in the existing simulation literature.

By working with existing simulation datasets, this project shows that meaningful molecular visualizations of LNP behavior are within reach even at the undergraduate level. These findings have direct relevance to biopharmaceuticals, providing a clearer mechanistic understanding of how lipid composition governs endosomal escape. I hope to accelerate the rational design of safer, more efficient mRNA delivery systems for both therapeutic and preventative applications.

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